Module 2

Accounting Statements and Finance

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Why Do We Need Accounting?

1. What do firms own?
   - Real assets (tangible, intangible) and financial assets (cash, securities)

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   - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?
3. How much money did firms make?
   - Income? Cash flow?

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?

CNBC, August 18, 2026 (3:50),
"Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

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- Is Anthropic profitable in Q2 2026?
- We will have a clear understanding after this module

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Three Financial Statements

- **Balance sheet**
  - What do firms own and owe at a point in time

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  - How much did firms earn in the past year

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  - What do firms own and owe at a point in time
- **Income statement**
  - How much did firms earn in the past year
- **Statement of cash flows**
  - Cash inflows and outflows during the past year

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- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure

sec.gov/edgar/search

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SEC.gov | EDGAR

Electronic Data Gathering, Analysis, and Retrieval

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- Distinguish between market and book values
- Explain why income differs from cash flow
- Understand the taxation of corporate and personal income

Module 2, Part 1

The Balance Sheet

Balance Sheet

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$$\textbf{Assets} = \textbf{Liabilities} + \textbf{Equity}$$

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Example, Issuing Stock to Build a Data Center ($ Billions)

1. Today			
Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	**1**	**Total liabilities and equity**	**1**

Example, Issuing Stock to Build a Data Center ($ Billions)

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Breaking Down the Balance Sheet of a Business ($ Millions)

Current assets

Assets that can generate return within one year

Assets	
Current assets	
Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100
Fixed assets	
Tangible fixed assets	
Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500
Intangible asset	700
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Obligations due within a year

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Most liquid → Least liquid

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Paid first → Paid last

- **Liquidity**, the ease of converting an asset into cash

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- In general, book $\neq$ market value

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Book Value vs. Market Value

1. Book value of equity = Book value of assets − Book value of liabilities
   - E.g., Apple at the end of fiscal year 2025,
     - Book value of assets, \$359,241 Mn
     - Book value of liabilities, \$285,508 Mn
     - What is the book value of equity?

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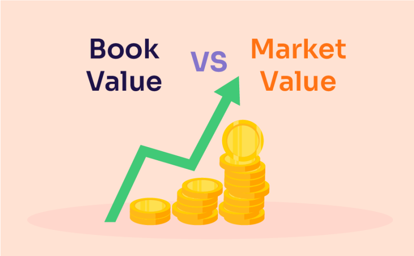
Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm



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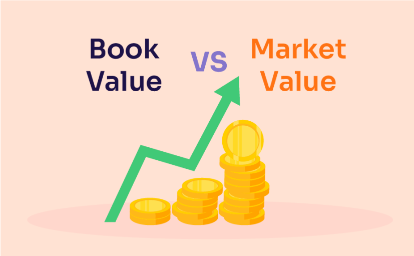

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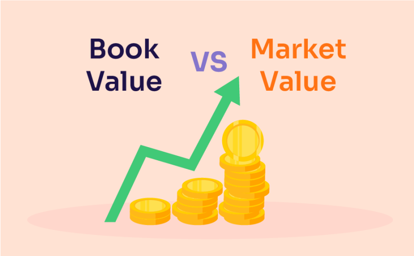
- Naturally, market value of equity $>$ book value of equity for a successful business



Practice Problem

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- Which of the following are current assets? Select all that apply.

a. Cash.
b. A software license (3-year term).
c. Wages payable (due in 2 weeks).
d. A bank loan (due in 3 years).
e. Accounts receivable.
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Recap of Part 1

- Accounting conveys information to investors and stakeholders
- The balance sheet, assets, liabilities, and equity at a point in time
- Assets $=$ Liabilities $+$ Equity
- In general, book $\neq$ market value

Module 2, Part 2

The Income Statement

Income Statement

- **Income statement**, how much firms earned in the past year

Income Statement

- **Revenues**, sales made in the period, whether or not cash was collected
  - E.g., a firm sells 100 products to customers, each at a price of \$100
  - Revenues $= 100 \times \$100 =$ \$10,000

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	**5,000**
Interest expense	1,000
Taxable income	**4,000**
Taxes	840
Net income	**3,160**
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Operating expenses**, the costs of running the business

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Income Statement

- **Operating expenses**, the costs of running the business
  - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
    - E.g., each of the 100 products sold costs the firm \$60 to make
    - COGS $= 100 \times \$60 =$ \$6,000

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- This spreading is **depreciation**
- Two things happen with depreciation,
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  - In the **balance sheet**, add to accumulated depreciation

Fixed Assets Spread over Time as Depreciation

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Effect on net income	0	**−0.4**	**−0.4**	**−0.4**	**−0.4**	**−0.4**	← Depreciation

Income Statement

- **EBIT**, earnings before interest and taxes
  - EBIT = Revenues − COGS − SG&A expenses − Depreciation

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Income Statement

- **Interest expense**, the interest paid on the firm's debt for the period

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Income Statement

- **Taxes**
  - Taxable income = EBIT − Interest expense
  - Taxes = Taxable income × Tax rate

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Income Statement

- **Net income**, what the business earned in the period after every expense

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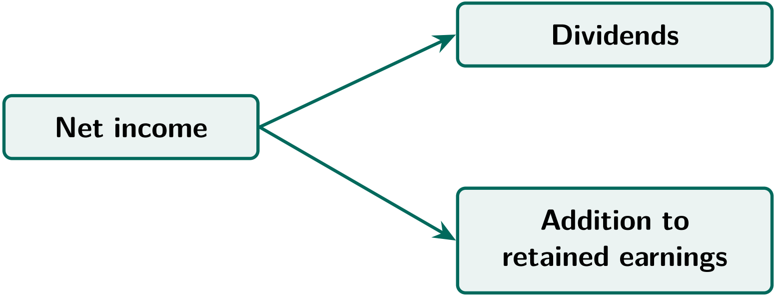

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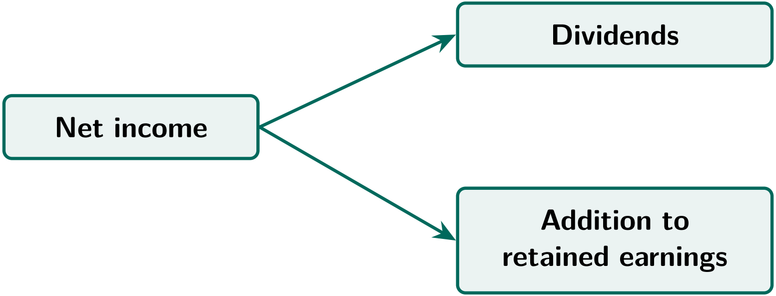

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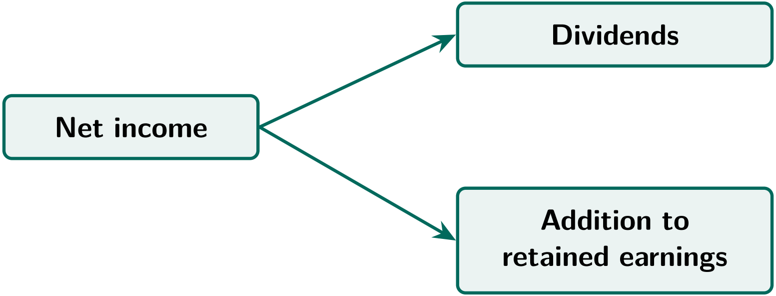

- **Dividends**, cash paid out to shareholders now
- **Retained earnings**, accumulated profits kept in the business and reinvested on the shareholders' behalf

Income Statement, A Memory Aid

Line item	Letters	Remember it as
Revenues	R	**R**ainbow
COGS	C	**C**andy
SG&A expenses	S	**S**ells
Depreciation	D	**D**ependably
EBIT	E	**E**specially
Interest expense	I	**I**n
Taxable income	T.I.	**T**exas **I**ncorporated
Taxes	T	**T**exas
Net income	N.I.	**N**ever **I**ncorporates
Dividends	D	**D**uring
Addition to retained earnings	R.E.	**R**e-**E**lections

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?

CNBC, August 18, 2026 (3:50),
"Anthropic tells investors annualized revenue run rate climbed to $65 billion in July"

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
  - EBITDA, earnings before interest, taxes, depreciation, and amortization
  - Amortization, the same cost-spreading as depreciation, applied to intangible assets

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- Not profitable on a **net income** basis

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Practice Problem

	$
Revenues	?
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	**12,000**
Interest expense	?
Taxable income	**?**
Taxes (tax rate = 21%)	2,100
Net income	**7,900**
Dividends	2,500
Addition to retained earnings	?

Practice Problem

	$
Revenues	50,000
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	**12,000**
Interest expense	2,000
Taxable income	**10,000**
Taxes (tax rate = 21%)	2,100
Net income	**7,900**
Dividends	2,500
Addition to retained earnings	5,400

Recap of Part 2

- The income statement, how much firms earned in the past year
- **Depreciation**, this period's share of a fixed asset's cost
- Net income can be kept in the business, or paid out to shareholders
- Taxes are computed **after** depreciation and interest, never before

Module 2, Part 3

The Statement of Cash Flows

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  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Statement of Cash Flows

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	**The common choice**

Direct →

Indirect →

Statement of Cash Flows

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- Both methods give the same totals

Direct →

Indirect →

Statement of Cash Flows

Direct method	Indirect method
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More time-consuming to prepare	Faster to prepare
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- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →

Indirect →

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Relationship of cash flow (CF) and net income (NI)?
When spending on fixed assets, CF = NI − spending on fixed assets
When recording depreciation, CF = NI + depreciation

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When raising money, CF = NI + amount raised

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- None of them have been sold yet

OpenAI, July 15, 2026 (2:15), "Introducing the Codex Micro"

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Cash flow (\$ millions)	**−60**
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Change in inventories	**+60**

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When firm not yet paid by customers, CF = NI − change in accounts receivable

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When firm pays off suppliers, CF = NI + change in accounts payable

Operating Activities

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
   - NWC = inventories + accounts receivable − accounts payable
   - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

When firm pays off suppliers, CF = NI + change in accounts payable

Why Subtract Change in Net Working Capital (NWC)?

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Ignoring other adjustments,

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, CF = NI − change in inventories

When firm not yet paid by customers, CF = NI − change in accounts receivable

When firm pays off suppliers, CF = NI + change in accounts payable

Ignoring other adjustments,

$$\begin{aligned} \textit{CF} = \textit{NI} & \\ & - \textit{change in inventories} \\ & - \textit{change in accounts receivable} \\ & + \textit{change in accounts payable} \end{aligned}$$

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$$\textit{CF} = \textit{NI} - \textit{change in net working capital (NWC)}$$

Operating Activities

Cash flow from operating activities	
Net income	3,160
Depreciation	3,000
Change in net working capital	−1,000
Cash flow from operating activities	7,160

$$3{,}160 + 3{,}000 - (-1{,}000) = 7{,}160$$

Investing Activities

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold
   - **Capital**, a firm's fixed assets
   - **Capital expenditure**, the amount spent on buying capital during the period
   - Cash flow is *negative* if firms buy capital

Investing Activities

Cash flow from investing activities	
Cash flow from capital expenditure	−3,000
Cash flow from sales of fixed assets	200
Cash flow from investing activities	−2,800

$$-3{,}000 + 200 = -2{,}800$$

Financing Activities

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Financing activities show cash flows from,

1. Selling new shares $\rightarrow$ Cash $\uparrow$
2. Paying dividends $\rightarrow$ Cash $\downarrow$
3. Taking new debt, either a loan or a bond $\rightarrow$ Cash $\uparrow$
4. Repaying principal amount of old debt $\rightarrow$ Cash $\downarrow$

Practice Problem

- During the year, a business sells new shares for \$120,000 and pays dividends of \$30,000. It takes a new loan of \$80,000 and repays \$50,000 of principal on old debt. How much is its cash flow from financing activities?

Practice Problem

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- Cash flow from financing activities $= \$120{,}000 - \$30{,}000 + \$80{,}000 - \$50{,}000 = \$120{,}000$

Recap of Part 3

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
  - **Operating activities**, cash flow $=$ Net income $+$ Depreciation $-$ Change in NWC
  - **Investing activities**, cash spent buying fixed assets and cash received from selling them
  - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Practice Problem

- Which of the following will make cash flow higher than net income for a firm?

a. Increase in inventories.
b. Decrease in accounts payable.
c. Decrease in accounts receivable.
d. Expenditures on fixed assets.

Practice Problem

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- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by $-$\$1,000 million. What is its cash flow from operating activities?

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Practice Problem

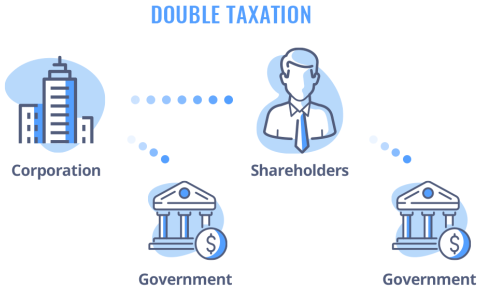
- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by $-$\$1,000 million. What is its cash flow from operating activities?
- Cash flow from operating activities = Net income + Depreciation − Change in NWC
- $\$3{,}160 + \$3{,}000 - (-\$1{,}000) = \$7{,}160$ million

Module 2, Part 4

Taxation

Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income



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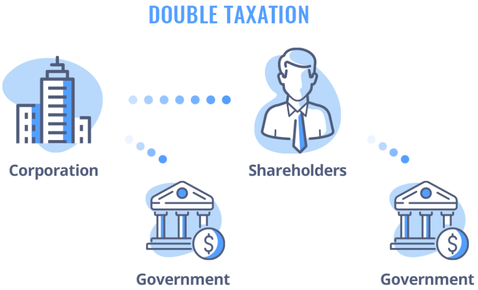
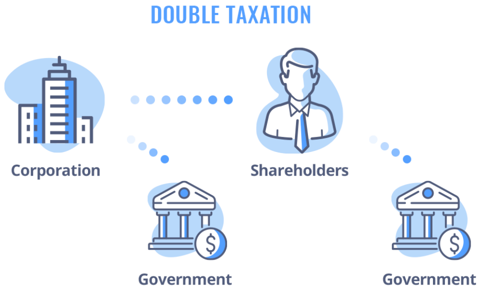


Corporate Taxation, Double Taxation

- Recall from **Module 1**,
  - The corporation pays tax on its income
  - Shareholders pay tax again on what they receive
- This is **double taxation**



Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**

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Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **Tax Cuts and Jobs Act (TCJA, 2017)**, cut the rate from 35% to 21%
- The tax cut was extended and largely made permanent in 2025

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Interest Tax Shield

- Interest comes first

Line item	Letters	Remember it as
Revenues	R	**R**ainbow
COGS	C	**C**andy
SG&A expenses	S	**S**ells
Depreciation	D	**D**ependably
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Interest expense	I	**I**n
Taxable income	T.I.	**T**exas **I**ncorporated
Taxes	T	**T**exas
Net income	N.I.	**N**ever **I**ncorporates
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- Interest comes first
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- Interest reduces corporate taxes

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Interest Tax Shield

Firm A	
EBIT	**$5,000**
Interest expense	$1,000
Taxable income	**?**
Taxes (tax rate = 21%)	?
Net income	**?**

Firm B	
EBIT	**$5,000**
Interest expense	$0
Taxable income	**?**
Taxes (tax rate = 21%)	?
Net income	**?**

Interest Tax Shield

Firm A	
EBIT	**\$5,000**
Interest expense	\$1,000
Taxable income	**\$4,000**
Taxes (tax rate = 21%)	\$840
Net income	**\$3,160**

Firm B	
EBIT	**\$5,000**
Interest expense	\$0
Taxable income	**?**
Taxes (tax rate = 21%)	?
Net income	**?**

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Firm B	
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Interest expense	$0
Taxable income	**$5,000**
Taxes (tax rate = 21%)	$1,050
Net income	**$3,950**

Interest Tax Shield

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EBIT	**$5,000**	
Interest expense	$0	$\rightarrow$ debt investors
Taxable income	**$5,000**	
Taxes (tax rate = 21%)	$1,050	
Net income	**$3,950**	

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Net income	**\$3,950** $\rightarrow$ equity investors

\$3,950

Investors get \$4,160 in total with debt, \$3,950 without

Interest Tax Shield

- Interest expense can
  - act as a "shield" against corporate taxation
  - increase the *total* cash flow to investors (debt and equity combined)

- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

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- In the example, $21\% \times \$1{,}000 = \$210$

Math for Interest Tax Shield

- Each dollar of interest lowers taxable income by $1

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- So taxes fall by the tax rate $\times$ \$1

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$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate})\times\$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\downarrow \text{ by tax rate}\times\$1}$$

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- Each dollar of interest raises the *total* cash flow to investors by the tax rate × \$1
- Should a firm take on as much debt as possible? No, bankruptcy costs.

Practice Problem

- A business pays a 21% tax rate. It takes on debt and pays \$1 more in interest each year. In that year, how does the cash flow to each investor group change?

a. Debt investors get \$1 more, equity investors get \$1 less, no change in total
b. Debt investors get \$1 more, equity investors get \$0.79 less, \$0.21 more in total
c. Debt investors get \$0.79 more, equity investors get \$1 less, \$0.21 less in total
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d. Debt investors get \$1 more, equity investors get \$0.21 less, \$0.79 more in total

- Net income falls by $(1 - 21\%) \times \$1 = \$0.79$, and the interest tax shield is $21\% \times \$1 = \$0.21$

Depreciation Tax Shield

- Depreciation comes first

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Revenues	R	**R**ainbow
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Depreciation Tax Shield

- Example, Suppose a business has an *additional* $1,000 million in depreciation, and its tax rate is 21%

Income Statement ($ Million)

	Baseline	
Revenues	80,000	
COGS	56,000	
SG&A expenses	16,000	
Depreciation	3,000	
EBIT	**5,000**	
Interest expense	1,000	
Taxable income	**4,000**	
Taxes	840	
Net income	**3,160**	

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Depreciation	3,000	4,000
EBIT	**5,000**	**4,000**
Interest expense	1,000	1,000
Taxable income	**4,000**	**3,000**
Taxes	840	630
Net income	**3,160**	

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Interest expense	1,000	1,000
Taxable income	**4,000**	**3,000**
Taxes	840	630
Net income	**3,160**	**2,370**

Depreciation Tax Shield

- Net income *falls* by $790 million

$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	−790
Add back depreciation			
Cash flow			

Depreciation Tax Shield

- Net income *falls* by \$790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation − Change in NWC

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	−790
Add back depreciation	3,000	4,000	+1,000
Cash flow			

Depreciation Tax Shield

- Net income *falls* by \$790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation − Change in NWC
- Equity investors have claims to cash flow from operating activities

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Net income	3,160	2,370	−790
Add back depreciation	3,000	4,000	+1,000
Cash flow			

Depreciation Tax Shield

- Net income *falls* by \$790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation − Change in NWC
- Equity investors have claims to cash flow from operating activities
- Cash flow *rises* by \$210 million

\$ Million	**Baseline**	**More depreciation**	**Change**
Net income	3,160	2,370	−790
Add back depreciation	3,000	4,000	+1,000
Cash flow	**6,160**	**6,370**	**+210**

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- Depreciation can also
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- Depreciation reduces taxable income, cuts taxes, and raises cash flow

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- Each dollar of depreciation raises the *total* cash flow to investors by the tax rate $\times$ \$1

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Depreciation tax shield = Tax rate × Depreciation

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- Equity investors get tax rate × \$1 more
- Each dollar of depreciation raises the *total* cash flow to investors by the tax rate × \$1
- Can a firm have as much depreciation as possible? No, constrained by cost of acquisition and accounting rules.

Practice Problem

- A business pays a 21% tax rate. Suppose it had an additional $200 million in depreciation. By how much would its taxes fall? What about its cash flow?

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- Taxes fall by $21\% \times \$200$ million $=$ \$42 million
- Cash flow rises by the same \$42 million, the depreciation tax shield

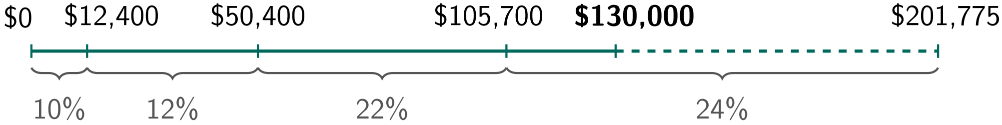
Personal Taxation

- Personal tax rates matter for salaried employees and non-corporate businesses
- **Marginal tax rate**, the tax rate applied to each extra dollar of income
- **Average tax rate**, total taxes / total income

Personal Taxation

Taxable Income (Single Taxpayers, 2026)	Marginal Tax Rate (%)
$0–$12,400	10
$12,401–$50,400	12
$50,401–$105,700	22
$105,701–$201,775	24
$201,776–$256,225	32
$256,226–$640,600	35
$640,601 and above	37

Source, IRS Rev. Proc. 2025-32



Example, Average Tax Rate

Suppose your salary is \$130,000 per year. What would be your average tax rate?

Taxable Income (Single Taxpayers, 2026)	Marginal Tax Rate (%)
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- Total tax $= \$12{,}400 \times 10\% + \$(50{,}400 - 12{,}400) \times 12\% + \$(105{,}700 - 50{,}400) \times 22\% + \$(130{,}000 - 105{,}700) \times 24\% = \$23{,}798$
- Average tax rate $= 23{,}798/130{,}000 = 18.31\%$

Practice Problem

- Suppose your salary is $30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

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- Marginal tax rate $= 12\%$

Recap of Part 4

- The U.S. federal corporate tax rate is **21%**
- Interest and depreciation are deducted (subtracted) before taxes, so both shield income from tax
- A tax shield is worth the tax rate $\times$ the deduction, raising the *total* cash flow to investors
- Marginal rate applies to each extra dollar of income, average rate is total taxes over total income

Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved

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- Net income $\neq$ cash flow, add back depreciation, subtract the change in NWC
- Interest and depreciation shield taxes, raising the *total* cash flow to investors

Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?

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- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900

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- EBIT = Taxable income + Interest expense = \$10,000 + \$2,000 = \$12,000

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- Taxable income = Taxes / Tax rate = \$2,100 / 21% = \$10,000
- Net income = Taxable income − Taxes = \$10,000 − \$2,100 = \$7,900
- EBIT = Taxable income + Interest expense = \$10,000 + \$2,000 = \$12,000
- Revenues = EBIT + COGS + SG&A expenses + Depreciation = \$12,000 + \$20,000 + \$10,000 + \$8,000 = \$50,000

Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

a. Inventories decrease while accounts receivable and accounts payable stay the same.
b. Accounts receivable increases while inventories and accounts payable stay the same.
c. The firm raises money by issuing stocks.
d. A large addition is made to plant and equipment.

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Thank you!

Please let me know if you have any questions.